

# LE HONG PHAT

## Data Engineer Intern

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[Github](#) | [Portfolio](#) | [Linkedin](#)

## EDUCATION

### Saigon University

Bachelor of Information Technology

Sep 2022 – Expected Mar 2027

Ho Chi Minh City, Vietnam

- Academic Excellence Scholarship - Awarded for 3 consecutive semesters
- Achieved Highest Semester with GPA: 8.8/10

### Certifications

- IBM Data Engineering (Aug 2026)
- IBM Data Analyst (Feb 2025)
- SQL Advanced — HackerRank (Dec 2025)

Languages: Vietnamese (Native), English (B2, IELTS in progress)

## TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

Data Engineering: dbt, Apache Kafka, Debezium, Apache Airflow

Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Redis

Tools & Backend: Docker, Git, FastAPI, SQLAlchemy, Prometheus, Grafana

Concepts: ETL/ELT, Data Modeling, Data Warehousing, Batch/Stream Processing, Medallion Architecture

## WORK EXPERIENCE

### Data Engineer Intern — AFFINA Insurance

Sep 2025 – May 2026

Ho Chi Minh City

- Implemented an end-to-end CDC pipeline using Debezium, capturing changes from 9 MySQL source tables and processing them into 10 staging tables, alongside weekly Excel batch uploads for downstream reporting.
- Developed a schema standardization pipeline using Factory, Strategy, and Template Method patterns to normalize heterogeneous insurance data, including column mappings, dates, monetary values, business keys, and domain-specific fields into a canonical contract schema.
- Implemented Redis-based cross-channel deduplication using 7 normalized business keys, enforcing an “online wins” policy to prevent duplicate offline uploads from conflicting with CDC records.

Tech: Python, FastAPI, MySQL, Debezium, Redis, Docker

## PROJECTS

### Hybrid Data Ingestion & Streaming Platform

Jun 2026

[github.com/phatle224/hybrid-data-ingestion-streaming-platform](https://github.com/phatle224/hybrid-data-ingestion-streaming-platform)

Independent implementation inspired by data-ingestion patterns encountered during my internship.

- Built a hybrid ELT platform combining CDC (Debezium + Kafka) and batch ingestion into PostgreSQL for near real-time analytics.
- Designed a four-layer dbt pipeline: Staging → Intermediate → Warehouse → Mart with 21 models and 101 configured data-quality tests, including five dimension and two fact models.
- Implemented idempotent deduplication using composite business keys and ROW\_NUMBER() window functions to eliminate duplicate records from CDC streams.

Tech: Python, Debezium, Apache Kafka, dbt, PostgreSQL, Prometheus, Grafana

### InsightMesh — Privacy-Bounded NL2SQL Analytics Platform

Sep 2026

[github.com/phatle224/insightmesh-multi-source-analytics](https://github.com/phatle224/insightmesh-multi-source-analytics)

- Built a local-first analytics workspace that enables users to connect read-only PostgreSQL and MySQL sources, explore schema relationships, ask questions in natural language.
- Engineered a deterministic FastAPI query pipeline combining hybrid schema retrieval, dialect-aware SQL generation, SQLGlot validation, bounded repair, read-only execution, and result verification while keeping raw rows and credentials out of LLM context by default.
- Developed a Next.js and TypeScript interface for datasource onboarding, interactive ERD exploration, execution traces, charts, and dashboards; benchmarked 37 PostgreSQL/MySQL cases on a 6-table demo schema, recording 100% result and join-path accuracy, 66.98% mean entity precision, and 5/5 unsafe cases blocked.

Tech: Python, FastAPI, PostgreSQL, MySQL, pgvector, SQLAlchemy, SQLGlot, Next.js, React, TypeScript